

ABOUT ME

I am an applied microeconomist in training, with research focused on rural credit markets and consumer decision making through experimental methods. My independent projects include causal inference applying quasi-experimental methods and econometric modeling to high frequency observational data to evaluate interventions and inform policy.

EDUCATION

University of California, Davis Davis, California
Ph.D. Candidate, Department of Agricultural and Resource Economics Sep 2021–Jun 2027 (Expected)

- Fields: Applied Microeconomics, Development Economics, Credit Markets, Behavioral Economics
- Dissertation: Repayment Norms and Credit Rationing in Microfinance
 - Causal identification of repayment norms through a framed field experiment with microfinance borrowers
 - Measurement of individual-specific norm sensitivity and its relationship to risk rationing and insurance take-up
 - Impact evaluation of risk-mitigation tools on credit rationing in microfinance through a randomized controlled trial (forthcoming)
- Funding: Markets, Risk and Resilience (MRR) Innovation Lab at UC Davis sub-project grant of \$30,740
- Awards: ARE Summer Research Fellowship, 2024 and 2026; ARE Graduate Travel Award, 2025

University of Dhaka Dhaka, Bangladesh
Master of Social Sciences in Economics Jan 2019–Dec 2019

- Coursework: Microeconomic and Macroeconomic Theory, Advanced Econometrics
- Ranked in the top 2 percent; CGPA 3.88/4.00
- Award: “Shamsun Nahar Mahmud Foundation Merit Scholarship” in 2022

Bachelor of Social Sciences in Economics Jan 2015–Dec 2018

- Coursework: Real Analysis, Dynamic Optimization, Statistics and Econometrics (Panel, Nonlinear, Time Series)
- Ranked in the top 1 percent; CGPA 3.96/4.00
- Award: “Governor’s Scholarship” in 2015, 2016, and 2017

WORK EXPERIENCE

Graduate Student Researcher California, USA
Agricultural and Resource Economics Department, UC Davis Summer 2022–Present

- Designing and implementing a randomized controlled trial to evaluate the impact of index insurance on credit rationing among small-scale agricultural producers with Prof. Michael Carter.
- Conceptualizing and designing a framed field experiment to elicit repayment norms among microfinance borrowers.
- Conducted a field survey and designed questionnaires for *constraint elicitation among microfinance borrowers* as a preparatory analysis for the index insurance project.
- Coordinating communications with collaborating partners, including drafting progress reports and working papers.

Teaching Assistant California, USA
Agricultural and Resource Economics Department, UC Davis Fall 2021–Spring 2023

- Led and prepared teaching materials for weekly discussion sessions for Econometric Theory, Econometrics for Business Decisions, Financial Management, and Microeconomics (30–180 students).
- Managed grading of assignments and exams, and held weekly office hours to support student learning.

Research Associate (Study Leave) Bangladesh
BRAC Institute of Governance and Development (BIGD) Aug 2020–Aug 2021

- Led research protocol implementation in the Economic Growth and Development cluster (PI: Dr. Narayan Das).
- Contributed to an impact evaluation project enhancing skills of underprivileged youth.
- Developed questionnaires, trained enumerators, supervised surveys, cleaned and analyzed data, and contributed to writing research reports, including a chapter on youth migration with an analysis of characteristics of migrant households for the Urban Development Program.

Research Assistant Bangladesh
BRAC James P. Grant School of Public Health Feb 2020–Jul 2020

- Analyzed data and drafted reports on the Health System’s Response to COVID-19 in Bangladesh, and contributed a chapter on social return on investment with survey design and data visualization for BRAC’s Health, Nutrition, and Population Program with Dr. Zahidul Quayyum.

Bangladesh Institute of Development Studies (BIDS) Aug 2019–May 2020
• Analyzed price volatility indices, managed large datasets of agricultural commodity prices, and explored GIS techniques for a project on crop diversification trends in Bangladesh with Dr. Mohammad Mainul Hoque.

Department of Economics, University of Dhaka Sep 2019–Dec 2019
• Prepared report on gender budget for the Ministry of Health and Family Planning with Dr. Sayema Haque.

RESEARCH IN PROGRESS

Repayment Norms in Microfinance

This study examines how behavioral and normative incentives enforce repayment in uncollateralized microfinance lending. I design and run a framed lab-in-the-field experiment with 919 active borrowers in southern Bangladesh, varying the social observability of default to identify norms as an enforcement mechanism. Borrowers' willingness to forgo financial gains to avoid violating a norm measures its salience; I find that exposure to normative costs reduces voluntary default and can function as non-financial collateral.

Presentations: 2025 AAEA & WAEA Joint Annual Meeting; 14th Annual Bay Area Behavioral Economics and Experimental Workshop, 2025.

Does Insurance Reduce Risk Rationing in Microfinance?, with Michael Carter (ongoing project)

This study evaluates whether indexed financial tools can mitigate risk rationing among smallholder farmers in Bangladesh. In a large-scale randomized experiment, I analyze the impacts of standalone agricultural index insurance versus the insurance bundled with savings and contingent line of credit on different forms of credit constraints among microfinance borrowers. I hypothesize that risk rationing is more prevalent among borrowers with higher norm conformity, isolating the constraints imposed by "social collateral."

Eliciting Non-Price Rationing among Microfinance Borrowers, with Michael Carter, Steve Boucher

This study explores the prevalence of non-price credit rationing. Adapting the Direct Elicitation Method (DEM) for microfinance context in Bangladesh, I categorize borrowers into quantity-, price-, and risk-rationed groups.

Impact of Minimum Wage Policy on Consumer Demand: Evidence from US Fast Food Chains, with Modhurima Amin, Syed Badruddoza

This study uses a difference-in-differences design to estimate how fast-food minimum wage legislation affects store-level sales and customer traffic. Drawing on the policy's selective coverage, we compare affected large chains against unaffected competitors using a high-frequency panel of 680,000+ observations, with data processing and analysis implemented in Python.

When Corruption is the Norm: Can Signaling Environmental Compliance Reduce Bribery?, with Moogdho Mahzab, Nina Brooks, Aprajit Mahajan, Stephen Luby

The study investigates whether signaling regulatory compliance can reduce illicit payments in the structurally informal brick-kiln industry of Bangladesh. Using an instrumental-variables approach, we estimate the causal effect of displaying environmental compliance banners on the bribes and compliance-related payments to the local bureaucrats. Contrary to concerns that signaling might increase extortion, we find that compliance signaling reduce illicit costs.

Presentations: PacDev 2026 (presented by co-author); AEDSB Dhaka Winter Conference, BIGD-BRAC (2025).

ADDITIONAL CONFERENCE PRESENTATIONS & AWARDS

- "Who Benefits from the Agricultural Microcredit Program?"—A preliminary theoretical analysis of risk rationing among agricultural microfinance borrowers. (May 2023)
 - Presented at *Fourth SANEM-World Bank North America Discussion Forum 2023*, Washington, DC
- "Environmental Kuznets Curve Analysis: Evidence from South Asia"—Panel cointegration analysis of pollutants and economic growth in seven South Asian countries (2019).
 - Best Paper Award, *15th South Asian Economics Students' Meet*, World Bank, Colombo, Sri Lanka
 - Finalist, *Budding Economist* competition—one of six undergraduate finalists from South Asia
 - Presented at *Session for Budding Researchers*, 4th SANEM Annual Economists' Conference, Dhaka, Bangladesh
- Best Paper Award, *1st Bangladesh Economics Summit*, University of Dhaka; "Higher Education Attainment and Socioeconomic Disparity in Bangladesh"—Cross-sectional logistic regression analysis using BDHS 2014 (2018).
- Best Paper Award, *Speakonomics at EconFest 4.0*, North South University; "Can Higher Education Reduce the Job Market Skill Gap?"—Cross-sectional analysis using the Bangladesh Labor Force Survey (2018).

SKILLS & METHODS

- **Programming:** Python (pandas, scikit-learn, statsmodels, data pipelines, string-matching), Stata, R, Julia, SQL
- **Statistical Methods:** Causal inference (DiD, IV, Event Study, SC), Experimental design, Panel data econometrics, ML (regression, neural networks, decision trees, gradient boosting, random forests), NLP
- **Tools & Platforms:** VS Code, L^AT_EX, Git, SurveyCTO, ArcGIS
- **Languages:** English, Bangla (native); Spanish, Hindi (conversational)

CERTIFICATION COURSES

- *Advanced Learning Algorithms*, Stanford & DeepLearning.AI, Coursera: neural networks, decision trees, ensemble methods, and model evaluation. (May 2026)
- *Supervised Machine Learning: Regression and Classification*, Stanford & DeepLearning.AI, Coursera: linear and logistic regression, regularization, and gradient descent. (Mar 2026)
- *Text Analysis and Natural Language Processing for Data Science*, DataLab, UC Davis: text preprocessing, topic modeling, and sentiment analysis. (Jun 2024)